

ONEFOUR, AB, Canada

WMO: 711160

Lat: 49.1231N

Lon: 110.4689W

Elev: 935

StdP: 90.59

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-25.8	-32.5	0.2	-29.1	-28.8	0.3	-25.6	17.9	-8.0	16.1	-6.8	4.9	310	0.751

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	33.2	16.4	31.2	15.9	29.0	15.5	18.4	27.7	17.4	27.0	16.5	26.1	4.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.6	12.4	20.5	14.3	11.4	19.7	13.1	10.6	18.8	55.8	27.5	52.5	27.0	49.6	26.0	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.2	12.4	11.0	DB	-32.5	36.3	2.9	1.7	-34.6	37.5	-36.3	38.5	-37.9	39.5	-40.1	40.8	(4)
(5)			WB	-32.5	20.2	2.9	1.4	-34.6	21.2	-36.3	22.1	-38.0	22.9	-40.1	23.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-8.8	-7.4	-1.4	5.4	11.0	15.5	20.2	19.3	13.7	6.1	-1.6	-7.7	(6)	
	DBStd	11.95	8.83	8.43	7.77	5.24	4.49	3.62	3.52	3.80	4.91	5.33	7.29	8.11	(7)	
	HDD10.0	2680	583	487	356	151	41	3	0	0	19	141	350	549	(8)	
	HDD18.3	4885	841	720	613	388	229	98	21	36	153	380	598	807	(9)	
	CDD10.0	1010	0	0	1	14	73	169	315	287	129	20	2	0	(10)	
	CDD18.3	173	0	0	0	0	2	14	78	65	13	0	0	0	(11)	
	CDH23.3	2598	0	0	0	6	80	236	1103	917	247	9	0	0	(12)	
	CDH26.7	1029	0	0	0	1	16	68	477	383	85	0	0	0	(13)	
(14) Wind	WSAvg	5.2	5.4	5.1	5.4	5.7	5.6	5.2	4.9	4.8	4.8	5.2	5.0	5.0	(14)	
(15) Precipitation	PrecAvg	347	26	16	25	25	50	58	32	35	33	14	15	25	(15)	
	PrecMax	555	68	73	105	88	143	202	71	83	182	42	43	70	(16)	
	PrecMin	176	6	1	2	0	5	2	1	3	6	0	1	5	(17)	
	PrecStd	93	20	15	20	19	33	43	21	20	38	12	13	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	12.3	18.2	23.9	28.5	32.1	35.8	35.3	32.3	24.7	17.5	10.1	(19)	
		MCWB	5.4	5.6	8.8	11.3	14.2	16.6	17.3	16.6	15.1	12.7	9.1	4.4	(20)	
	2%	DB	7.5	8.3	14.5	20.3	25.6	28.4	33.7	32.8	29.2	20.5	13.5	7.2	(21)	
		MCWB	3.5	3.4	6.9	10.1	13.3	15.5	16.9	15.7	14.2	10.8	6.7	2.9	(22)	
	5%	DB	5.3	5.7	11.5	17.2	22.9	25.9	31.6	30.9	26.3	17.4	10.5	5.1	(23)	
		MCWB	2.3	1.9	5.2	8.3	12.1	14.9	16.6	15.5	13.4	9.3	5.0	1.7	(24)	
	10%	DB	3.2	3.3	8.5	14.5	20.1	23.8	29.3	28.6	23.1	14.6	7.8	3.1	(25)	
		MCWB	0.8	0.6	3.6	6.9	10.8	14.1	16.2	15.2	12.3	8.0	3.5	0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	5.9	9.0	12.2	15.5	19.1	20.7	18.6	17.0	14.2	10.0	5.3	(27)	
		MCDB	9.8	11.8	17.1	21.3	25.1	26.7	29.9	28.0	27.7	19.2	15.8	8.9	(28)	
	2%	WB	3.8	3.7	7.2	10.5	14.1	17.2	19.0	17.5	15.3	11.4	7.2	3.3	(29)	
		MCDB	7.0	7.8	13.5	19.0	23.0	25.0	28.1	27.6	25.4	19.1	12.9	6.7	(30)	
	5%	WB	2.3	2.1	5.6	9.0	12.9	15.9	17.9	16.7	14.1	9.8	5.4	1.9	(31)	
		MCDB	5.2	5.4	10.9	16.6	21.4	23.6	27.9	27.0	23.8	16.5	10.0	4.9	(32)	
	10%	WB	0.8	0.7	3.9	7.4	11.6	14.7	17.0	15.8	12.9	8.3	3.7	0.4	(33)	
		MCDB	3.2	3.2	8.3	13.6	18.8	21.8	26.6	26.1	22.4	13.9	7.4	3.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	10.2	10.9	12.3	12.9	12.6	15.1	15.1	14.1	12.6	10.8	9.9	(35)	
		MCDBR	12.3	12.7	15.2	17.4	17.3	16.5	18.5	19.0	18.8	17.1	14.4	12.3	(36)	
		MCWBR	9.1	8.6	8.5	9.0	7.9	6.8	6.5	6.7	7.9	8.8	8.9	8.6	(37)	
	5% WB	MCDBR	12.5	12.5	14.7	16.6	15.5	14.6	15.9	16.0	17.1	15.9	13.8	12.0	(38)	
		MCWBR	9.4	8.9	8.7	8.8	7.4	6.7	6.4	6.2	7.6	8.6	8.8	8.7	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.264	0.262	0.285	0.334	0.349	0.345	0.346	0.366	0.322	0.290	0.261	0.247	(40)	
	taud		2.431	2.499	2.492	2.359	2.358	2.426	2.433	2.352	2.447	2.529	2.514	2.450	(41)	
	Ebn at Noon		815	913	939	915	910	914	907	870	882	862	819	792	(42)	
	Edh at Noon		62	73	88	112	117	110	108	112	92	71	56	53	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.29	2.26	3.55	4.92	5.86	6.54	7.11	5.73	4.09	2.56	1.46	1.01	(44)	
	RadStd		0.11	0.19	0.27	0.31	0.44	0.38	0.37	0.36	0.32	0.24	0.10	0.09	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page